

Laboratory Requisition — Trebuchet Investigation

Unit: Forces & Motion (Yr 7–8)

Requested

by: Date needed: _____

Lesson: Culminating investigation

Number of groups: _____

Period / room: _____

What the students are doing

Students launch a projectile horizontally from a trebuchet, measure the launch height (h) and the horizontal distance the projectile travels (R), and film each launch in slow motion to measure the time of flight. They use this to calculate gravity (g). Each group needs to run **10 launches**, so allow 50–60 minutes.

Equipment required

☐ Item	Per group	Total needed	Notes
☐ Trebuchet (desktop model)	1	_____	Class set, or shared between groups
☐ Projectile	1	_____	40 g ball bearing or golf ball; identical for all groups
☐ Spare projectiles	2	_____	Same mass — in case of loss
☐ Measuring tape (5 m+)	1	_____	For measuring range R
☐ Metre ruler	1	_____	For measuring launch height h
☐ Spirit level	1	_____	To check trebuchet is level
☐ Plumb line	1	_____	To mark the origin point on the floor
☐ Masking tape	1 roll	_____	Mark origin + exclusion zone
☐ Smartphone or tablet + tripod	1	_____	Must shoot slow motion (≥ 120 fps)
☐ Safety glasses	4	_____	One per student — mandatory

Setup before lesson

Safety

1. Place each trebuchet on a level bench or floor; check with spirit level.

2. Hang the plumb line from the projectile’s resting (release) position and mark the point directly below it on the floor with tape. Label it “0 m”.

3. Lay the measuring tape along the floor from the “0 m” mark toward the landing zone.

4. Mark a **1 m exclusion zone** around each trebuchet with tape.

5. Confirm each camera is charged and slow-motion mode is switched on.
- Risk level: Medium.** Teacher supervises all launches.

Safety glasses worn by all students during launching.

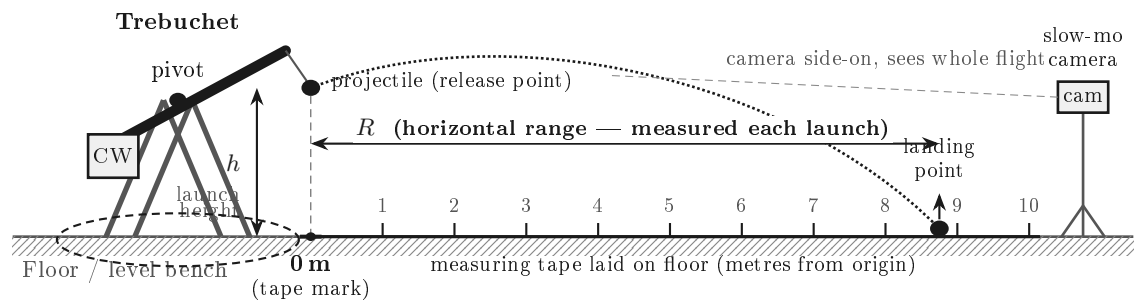
Only one person operates each trebuchet; others stand **behind** it.

Nobody in front of or beside the launch path.

Projectiles retrieved only on teacher’s “all clear”.

Arm fully reset before reloading.

Setup diagram (plan for one station)



Key: h = vertical height from projectile release point to floor (measured once, fixed). R = horizontal distance from the “0 m” origin to where the projectile lands (measured each launch). The camera films side-on so students can count video frames to find the time of flight.

Notes / queries for teacher: _____

Prepared by (tech): _____